

MYSQL TEST

1. List all customers ordered by customerName

Query :- select * from customers order by customerName;

Results :-

```
mysql> select * from customers order by customerName;
```

customerNumber	customerName	contactLastName	contactFirstName	phone	addressLine1	addressLine2	city	state	postalCode	country	salesRepEmployeeNumber	creditLimit
2881	ABC Corp	Lee	Chris	1111111111	Street 1	NULL	NYC	NY	10001	USA	1002	50000.00
2883	Global Inc	Singh	Raj	3333333333	Street 3	NULL	Delhi	NULL	110001	India	1002	60000.00
2882	XYZ Ltd	Kim	Pat	2222222222	Street 2	NULL	London	NULL	NW1	UK	1003	70000.00

3 rows in set (0.00 sec)

2. Display all products ordered by buyPrice descending

Query :- select * from products order by buyPrice DESC;

Results :-

```
mysql> select * from products order by buyPrice DESC;
```

productCode	productName	productLine	productScale	productVendor	productDescription	quantityInStock	buyPrice	MSRP
P101	Toyota Camry	Cars	1:18	Toyota	Sedan model	100	20.00	30.00
P102	Honda Civic	Cars	1:18	Honda	Compact model	150	18.00	28.00
P201	Yamaha Bike	Bikes	1:12	Yamaha	Sport bike	200	10.00	15.00

3 rows in set (0.00 sec)

3. Show all employees ordered by lastName

Query :- select * from employees order by lastName;

Results :-

```
mysql> select * from employees order by lastName;
```

employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle
1003	Brown	Mike	x103	mike@company.com	2	1001	Sales Rep
1002	Doe	Jane	x102	jane@company.com	1	1001	Sales Rep
1001	Smith	John	x101	john@company.com	1	NULL	Manager

3 rows in set (0.05 sec)

4. List all offices sorted by country

Query :- select * from offices order by country;

Results :-

```
mysql> select * from offices order by country;
```

officeCode	city	phone	addressLine1	addressLine2	state	country	postalCode	territory
2	London	9876543210	Baker St	NULL	NULL	UK	NW1	EMEA
1	New York	1234567890	5th Ave	NULL	NY	USA	10001	NA

2 rows in set (0.01 sec)

5. Get all orders sorted by orderDate descending

Query :- select * from orders order by orderDate desc;

Results :-

```
mysql> select * from orders order by orderDate desc;
+-----+-----+-----+-----+-----+-----+-----+
| orderNumber | orderDate | requiredDate | shippedDate | status | comments | customerNumber |
+-----+-----+-----+-----+-----+-----+-----+
| 3003 | 2024-03-01 | 2024-03-06 | NULL | Pending | NULL | 2003 |
| 3002 | 2024-02-01 | 2024-02-05 | 2024-02-04 | Shipped | NULL | 2002 |
| 3001 | 2024-01-01 | 2024-01-05 | 2024-01-03 | Shipped | NULL | 2001 |
+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.02 sec)
```

6. Show customer names with their sales representative

Query :- select c.customername, e.firstname, e.lastname from customers c join employees e on c.salesrepemployeenumber = e.employeenumber;

Results :-

```
mysql> select c.customername, e.firstname, e.lastname from customers c join employees e on c.salesrepemployeenumber = e.employeenumber;
+-----+-----+-----+
| customername | firstname | lastname |
+-----+-----+-----+
| ABC Corp | Jane | Doe |
| Global Inc | Jane | Doe |
| XYZ Ltd | Mike | Brown |
+-----+-----+-----+
3 rows in set (0.02 sec)
```

7. List all products with their product line

Query :- select productName, productLine from products;

Results :-

```
mysql> select productName, productLine from products;
+-----+-----+
| productName | productLine |
+-----+-----+
| Toyota Camry | Cars |
| Honda Civic | Cars |
| Yamaha Bike | Bikes |
+-----+-----+
3 rows in set (0.02 sec)
```

8. Display orders with customer names

Query :- select o.ordernumber, c.customername
from orders o
join customers c
on o.customernumber = c.customernumber;

Results :-

```
mysql> select o.ordernumber, c.customername
-> from orders o
-> join customers c
-> on o.customernumber = c.customernumber;
+-----+-----+
| ordernumber | customername |
+-----+-----+
| 3001 | ABC Corp |
| 3002 | XYZ Ltd |
| 3003 | Global Inc |
+-----+-----+
3 rows in set (0.00 sec)
```

9. Show order details with product names

Query :- select od.ordernumber, p.productname, od.quantityordered
from orderdetails od
join products p
on od.productcode = p.productcode;

Results :-

```
mysql> select od.ordernumber, p.productname, od.quantityordered
-> from orderdetails od
-> join products p
-> on od.productcode = p.productcode;
+-----+-----+-----+
| ordernumber | productname | quantityordered |
+-----+-----+-----+
| 3001 | Toyota Camry | 10 |
| 3001 | Yamaha Bike | 5 |
| 3002 | Honda Civic | 7 |
| 3003 | Toyota Camry | 3 |
+-----+-----+-----+
4 rows in set (0.02 sec)
```

10. List payments with customer names

Query :- select p.checknumber, c.customername, p.amount
from payments p
join customers c
on p.customernumber = c.customernumber;

Results :-

```
mysql> select p.checknumber, c.customername, p.amount
-> from payments p
-> join customers c
-> on p.customernumber = c.customernumber;
+-----+-----+-----+
| checknumber | customername | amount |
+-----+-----+-----+
| CHK001 | ABC Corp | 300.00 |
| CHK002 | XYZ Ltd | 500.00 |
| CHK003 | Global Inc | 150.00 |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

11. Count total number of customers

Query :- select count(*) as total_customers from customers;

Results :-

```
mysql> select count(*) as total_customers from customers;
+-----+
| total_customers |
+-----+
| 3 |
+-----+
1 row in set (0.02 sec)
```

12. Count total number of orders

Query :- select count(*) as total_orders from orders;

Results :-

```
mysql> select count(*) as total_orders from orders;
+-----+
| total_orders |
+-----+
|          3 |
+-----+
1 row in set (0.00 sec)
```

13. Find average buyPrice of products

Query :- select avg(buyprice) as avg_price from products;

Results :-

```
mysql> select avg(buyprice) as avg_price from products;
+-----+
| avg_price |
+-----+
| 16.000000 |
+-----+
1 row in set (0.00 sec)
```

14. Find maximum MSRP from products

Query :- select max(msrp) as max_price from products;

Results :-

```
mysql> select max(msrp) as max_price from products;
+-----+
| max_price |
+-----+
|      30.00 |
+-----+
1 row in set (0.02 sec)
```

15. Find total number of employees

Query :- select count(*) as total_employees from employees;

Results :-

```
mysql> select count(*) as total_employees from employees;
+-----+
| total_employees |
+-----+
|          3 |
+-----+
1 row in set (0.00 sec)
```

16. Count customers per country

Query :- select country, count(*) as total_customers
from customers
group by country;

Results :-

```
mysql> select country, count(*) as total_customers
-> from customers
-> group by country;
+-----+-----+
| country | total_customers |
+-----+-----+
| India   | 1               |
| UK      | 1               |
| USA     | 1               |
+-----+-----+
3 rows in set (0.02 sec)
```

17. Count orders per status

Query :- select productline, count(*) as total_products
from products
group by productline;

Results :-

```
mysql> select productline, count(*) as total_products
-> from products
-> group by productline;
+-----+-----+
| productline | total_products |
+-----+-----+
| Bikes       | 1              |
| Cars        | 2              |
+-----+-----+
2 rows in set (0.00 sec)
```

18. Count products per product line

Query :- select productline, count(*) as total_products
from products
group by productline;

Results :-

```
mysql> select productline, count(*) as total_products
-> from products
-> group by productline;
+-----+-----+
| productline | total_products |
+-----+-----+
| Bikes       | 1              |
| Cars        | 2              |
+-----+-----+
2 rows in set (0.00 sec)
```

19. Find average credit limit per country

Query :- select country, avg(creditlimit) as avg_credit
from customers
group by country;

Results :-

```
mysql> select country, avg(creditlimit) as avg_credit
-> from customers
-> group by country;
+-----+-----+
| country | avg_credit |
+-----+-----+
| India   | 60000.000000 |
| UK      | 70000.000000 |
| USA     | 50000.000000 |
+-----+-----+
3 rows in set (0.02 sec)
```

20. Count employees per office

Query :- select officecode, count(*) as total_employees
from employees
group by officecode;

Results :-

```
mysql> select officecode, count(*) as total_employees
-> from employees
-> group by officecode;
+-----+-----+
| officecode | total_employees |
+-----+-----+
| 1          | 2               |
| 2          | 1               |
+-----+-----+
2 rows in set (0.02 sec)
```

21. count number of orders per customer

Query :- select customerNumber, count(*) as total_orders
from orders
group by customerNumber;

Results :-

```
mysql> select customerNumber, count(*) as total_orders
-> from orders
-> group by customerNumber;
+-----+-----+
| customerNumber | total_orders |
+-----+-----+
| 2001          | 1           |
| 2002          | 1           |
| 2003          | 1           |
+-----+-----+
3 rows in set (0.00 sec)
```

22. find total payment amount per customer

Query :- select customerNumber, sum(amount) as total_payment
from payments
group by customerNumber;

Results :-

```
mysql> select customerNumber, sum(amount) as total_payment
-> from payments
-> group by customerNumber;
```

customerNumber	total_payment
2001	300.00
2002	500.00
2003	150.00

3 rows in set (0.00 sec)

23. list total quantity ordered per product

Query :- select productCode, sum(quantityOrdered) as total_quantity
from orderdetails
group by productCode;

Results :-

```
mysql> select productCode, sum(quantityOrdered) as total_quantity
-> from orderdetails
-> group by productCode;
```

productCode	total_quantity
P101	13
P102	7
P201	5

3 rows in set (0.00 sec)

24. show total revenue per product

Query :- select productCode, sum(quantityOrdered * priceEach) as total_revenue
from orderdetails
group by productCode;

Results :-

```
mysql> select productCode, sum(quantityOrdered * priceEach) as total_revenue
-> from orderdetails
-> group by productCode;
```

productCode	total_revenue
P101	390.00
P102	196.00
P201	75.00

3 rows in set (0.01 sec)

25. find number of customers handled by each employee

Query :- select salesRepEmployeeNumber, count(*) as total_customers
from customers
group by salesRepEmployeeNumber;

Results :-

```
mysql> select salesRepEmployeeNumber, count(*) as total_customers
-> from customers
-> group by salesRepEmployeeNumber;
```

salesRepEmployeeNumber	total_customers
1002	2
1003	1

```
2 rows in set (0.02 sec)
```

26. find customers with more than 5 orders

Query :- select customerNumber, count(*) as total_orders
from orders
group by customerNumber
having count(*) > 5;

Results :-

```
mysql> select customerNumber, count(*) as total_orders
-> from orders
-> group by customerNumber
-> having count(*) > 5;
Empty set (0.02 sec)
```

27. list products with total quantity ordered > 100

Query :- select productCode, sum(quantityOrdered) as total_quantity
from orderdetails
group by productCode
having sum(quantityOrdered) > 100;

Results :-

```
mysql> select productCode, sum(quantityOrdered) as total_quantity
-> from orderdetails
-> group by productCode
-> having sum(quantityOrdered) > 100;
Empty set (0.01 sec)
```

28. find employees managing more than 3 customers

Query :- select salesRepEmployeeNumber, count(*) as total_customers
from customers
group by salesRepEmployeeNumber
having count(*) > 3;

Results :-

```
mysql> select salesRepEmployeeNumber, count(*) as total_customers
-> from customers
-> group by salesRepEmployeeNumber
-> having count(*) > 3;
Empty set (0.00 sec)
```

29. show countries with more than 10 customers

Query :- select country, count(*) as total_customers
from customers
group by country
having count(*) > 10;

Results :-

```
mysql> select country, count(*) as total_customers
-> from customers
-> group by country
-> having count(*) > 10;
Empty set (0.00 sec)
```

30. find product lines having more than 5 products

Query :- select productLine, count(*) as total_products
from products
group by productLine
having count(*) > 5;

Results :-

```
mysql> select productLine, count(*) as total_products
-> from products
-> group by productLine
-> having count(*) > 5;
Empty set (0.00 sec)
```

31. list top 5 customers by total payment amount

Query :- select customerNumber, sum(amount) as total_payment
from payments
group by customerNumber
order by total_payment desc
limit 5;

Results :-

```
mysql> select customerNumber, sum(amount) as total_payment
-> from payments
-> group by customerNumber
-> order by total_payment desc
-> limit 5;
```

customerNumber	total_payment
2002	500.00
2001	300.00
2003	150.00

3 rows in set (0.02 sec)

32. show products ordered by total sales descending

Query :- select productCode, sum(quantityOrdered * priceEach) as total_sales
from orderdetails
group by productCode
order by total_sales desc;

Results :-

```
mysql> select productCode, sum(quantityOrdered * priceEach) as total_sales
-> from orderdetails
-> group by productCode
-> order by total_sales desc;
```

productCode	total_sales
P101	390.00
P102	196.00
P201	75.00

3 rows in set (0.02 sec)

33. list employees sorted by number of customers handled

Query :- select salesRepEmployeeNumber, count(*) as total_customers
from customers
group by salesRepEmployeeNumber
order by total_customers desc;

Results :-

```
mysql> select salesRepEmployeeNumber, count(*) as total_customers
-> from customers
-> group by salesRepEmployeeNumber
-> order by total_customers desc;
```

salesRepEmployeeNumber	total_customers
1002	2
1003	1

2 rows in set (0.00 sec)

34. show countries sorted by total customers

Query :- select country, count(*) as total_customers
from customers
group by country
order by total_customers desc;

Results :-

```
mysql> select country, count(*) as total_customers
-> from customers
-> group by country
-> order by total_customers desc;
+-----+-----+
| country | total_customers |
+-----+-----+
| USA     | 1               |
| UK      | 1               |
| India   | 1               |
+-----+-----+
3 rows in set (0.00 sec)
```

35. display top 10 most ordered products

Query :- select productCode, sum(quantityOrdered) as total_quantity
from orderdetails
group by productCode
order by total_quantity desc
limit 10;

Results :-

```
mysql> select productCode, sum(quantityOrdered) as total_quantity
-> from orderdetails
-> group by productCode
-> order by total_quantity desc
-> limit 10;
+-----+-----+
| productCode | total_quantity |
+-----+-----+
| P101        | 13             |
| P102        | 7              |
| P201        | 5              |
+-----+-----+
3 rows in set (0.00 sec)
```

36. get customer name, order number, and product name

Query :- select c.customerName, o.orderNumber, p.productName
from orders o
join customers c on o.customerNumber = c.customerNumber
join orderdetails od on o.orderNumber = od.orderNumber
join products p on od.productCode = p.productCode;

Results :-

```
mysql> select c.customerName, o.orderNumber, p.productName
-> from orders o
-> join customers c on o.customerNumber = c.customerNumber
-> join orderdetails od on o.orderNumber = od.orderNumber
-> join products p on od.productCode = p.productCode;
+-----+-----+-----+
| customerName | orderNumber | productName |
+-----+-----+-----+
| ABC Corp     | 3001        | Toyota Camry |
| Global Inc   | 3003        | Toyota Camry |
| XYZ Ltd      | 3002        | Honda Civic  |
| ABC Corp     | 3001        | Yamaha Bike  |
+-----+-----+-----+
4 rows in set (0.02 sec)
```

37. show employee name, customer name, and total payments

Query :- select e.firstName, e.lastName, c.customerName, sum(p.amount) as total_payment
from employees e
join customers c on e.employeeNumber = c.salesRepEmployeeNumber
join payments p on c.customerNumber = p.customerNumber
group by e.employeeNumber, c.customerNumber;

Results :-

```
mysql> select e.firstName, e.lastName, c.customerName, sum(p.amount) as total_payment
-> from employees e
-> join customers c on e.employeeNumber = c.salesRepEmployeeNumber
-> join payments p on c.customerNumber = p.customerNumber
-> group by e.employeeNumber, c.customerNumber;
+-----+-----+-----+-----+
| firstName | lastName | customerName | total_payment |
+-----+-----+-----+-----+
| Jane     | Doe     | ABC Corp     | 300.00        |
| Jane     | Doe     | Global Inc   | 150.00        |
| Mike     | Brown   | XYZ Ltd      | 500.00        |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

38. list order details with product line and quantity ordered

Query :- select od.orderNumber, p.productLine, od.quantityOrdered
from orderdetails od
join products p on od.productCode = p.productCode;

Results :-

```
mysql> select od.orderNumber, p.productLine, od.quantityOrdered
-> from orderdetails od
-> join products p on od.productCode = p.productCode;
```

orderNumber	productLine	quantityOrdered
3001	Cars	10
3001	Bikes	5
3002	Cars	7
3003	Cars	3

4 rows in set (0.00 sec)

39. display office city with employee count

Query :- select o.city, count(e.employeeNumber) as total_employees
from offices o
join employees e on o.officeCode = e.officeCode
group by o.city;

Results :-

```
mysql> select o.city, count(e.employeeNumber) as total_employees
-> from offices o
-> join employees e on o.officeCode = e.officeCode
-> group by o.city;
```

city	total_employees
London	1
New York	2

2 rows in set (0.00 sec)

40. show product vendor with total quantity sold

Query :- select p.productVendor, sum(od.quantityOrdered) as total_quantity
from products p
join orderdetails od on p.productCode = od.productCode
group by p.productVendor;

Results :-

```
mysql> select p.productVendor, sum(od.quantityOrdered) as total_quantity
-> from products p
-> join orderdetails od on p.productCode = od.productCode
-> group by p.productVendor;
```

productVendor	total_quantity
Honda	7
Toyota	13
Yamaha	5

3 rows in set (0.00 sec)

41. find average order value per customer

Query :- select o.customerNumber, avg(od.quantityOrdered * od.priceEach) as avg_order_value
from orders o
join orderdetails od on o.orderNumber = od.orderNumber
group by o.customerNumber;

Results :-

```
mysql> select o.customerNumber, avg(od.quantityOrdered * od.priceEach) as avg_order_value
-> from orders o
-> join orderdetails od on o.orderNumber = od.orderNumber
-> group by o.customerNumber;
+-----+-----+
| customerNumber | avg_order_value |
+-----+-----+
| 2001           | 187.500000     |
| 2002           | 196.000000     |
| 2003           | 90.000000      |
+-----+-----+
3 rows in set (0.00 sec)
```

42. show total revenue generated per country

Query :- select c.country, sum(od.quantityOrdered * od.priceEach) as total_revenue
from customers c
join orders o on c.customerNumber = o.customerNumber
join orderdetails od on o.orderNumber = od.orderNumber
group by c.country;

Results :-

```
mysql> select c.country, sum(od.quantityOrdered * od.priceEach) as total_revenue
-> from customers c
-> join orders o on c.customerNumber = o.customerNumber
-> join orderdetails od on o.orderNumber = od.orderNumber
-> group by c.country;
+-----+-----+
| country | total_revenue |
+-----+-----+
| India   | 90.00         |
| UK      | 196.00        |
| USA     | 375.00        |
+-----+-----+
3 rows in set (0.00 sec)
```

43. find customers whose total payments exceed 50000

Query :- select customerNumber, sum(amount) as total_payment
from payments
group by customerNumber
having sum(amount) > 50000;

Results :-

```
mysql> select customerNumber, sum(amount) as total_payment
-> from payments
-> group by customerNumber
-> having sum(amount) > 50000;
Empty set (0.00 sec)
```

44. list products where average order quantity > 50

Query :- select productCode, avg(quantityOrdered) as avg_qty
from orderdetails
group by productCode
having avg(quantityOrdered) > 50;

Results :-

```
mysql> select productCode, avg(quantityOrdered) as avg_qty  
-> from orderdetails  
-> group by productCode  
-> having avg(quantityOrdered) > 50;  
Empty set (0.00 sec)
```

45. find top 3 product lines by total revenue

Query :- select p.productLine, sum(od.quantityOrdered * od.priceEach) as total_revenue
from products p
join orderdetails od on p.productCode = od.productCode
group by p.productLine
order by total_revenue desc
limit 3;

Results :-

```
mysql> select p.productLine, sum(od.quantityOrdered * od.priceEach) as total_revenue  
-> from products p  
-> join orderdetails od on p.productCode = od.productCode  
-> group by p.productLine  
-> order by total_revenue desc  
-> limit 3;  
+-----+-----+  
| productLine | total_revenue |  
+-----+-----+  
| Cars        |          586.00 |  
| Bikes       |           75.00 |  
+-----+-----+  
2 rows in set (0.00 sec)
```

46. show customers with credit limit > average credit limit

Query :- select *
from customers
where creditLimit > (select avg(creditLimit) from customers);

Results :-

```
mysql> select *  
-> from customers  
-> where creditLimit > (select avg(creditLimit) from customers);  
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+  
| customerNumber | customerName | contactLastName | contactFirstName | phone | addressLine1 | addressLine2 | city | state | postalCode | country | salesRepEmployeeNumber | creditLimit |  
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+  
| 2002 | XYZ Ltd | Kim | Pat | 222222222 | Street 2 | NULL | London | NULL | NW1 | UK | 1003 | 70000.00 |  
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+  
1 row in set (0.01 sec)
```

47. find orders where total amount > 10000

Query :- select orderNumber, sum(quantityOrdered * priceEach) as total_amount
from orderdetails
group by orderNumber
having sum(quantityOrdered * priceEach) > 10000;

Results :-

```
mysql>
mysql> select orderNumber, sum(quantityOrdered * priceEach) as total_amount
      -> from orderdetails
      -> group by orderNumber
      -> having sum(quantityOrdered * priceEach) > 10000;
Empty set (0.00 sec)
```

48. list employees working in offices with more than 5 employees

Query :- select *
from employees
where officeCode in (
select officeCode
from employees
group by officeCode
having count(*) > 5
);

Results :-

```
mysql> select *
      -> from employees
      -> where officeCode in (
      -> select officeCode
      -> from employees
      -> group by officeCode
      -> having count(*) > 5
      -> );
Empty set (0.00 sec)
```

49. find products with stock less than average stock

Query :- select *
from products
where quantityInStock < (select avg(quantityInStock) from products);

Results :-

```
mysql> select *
      -> from products
      -> where quantityInStock < (select avg(quantityInStock) from products);
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| productCode | productName | productLine | productScale | productVendor | productDescription | quantityInStock | buyPrice | MSRP |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| P101        | Toyota Camry | Cars        | 1:18         | Toyota        | Sedan model        | 100             | 20.00   | 30.00 |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.01 sec)
```

50. show customers who made more than 3 payments

Query :- select customerNumber, count(*) as total_payments
from payments
group by customerNumber
having count(*) > 3;

Results :-

```
mysql> select customerNumber, count(*) as total_payments  
-> from payments  
-> group by customerNumber  
-> having count(*) > 3;  
Empty set (0.00 sec)
```

51. find total revenue generated by each customer

Query :- select c.customerName, sum(od.quantityOrdered * od.priceEach) as total_revenue
from customers c
join orders o on c.customerNumber = o.customerNumber
join orderdetails od on o.orderNumber = od.orderNumber
group by c.customerName
order by total_revenue desc;

Results :-

```
mysql> select c.customerName, sum(od.quantityOrdered * od.priceEach) as total_revenue  
-> from customers c  
-> join orders o on c.customerNumber = o.customerNumber  
-> join orderdetails od on o.orderNumber = od.orderNumber  
-> group by c.customerName  
-> order by total_revenue desc;  
+-----+-----+  
| customerName | total_revenue |  
+-----+-----+  
| ABC Corp    |          375.00 |  
| XYZ Ltd     |          196.00 |  
| Global Inc  |           90.00 |  
+-----+-----+  
3 rows in set (0.00 sec)
```

52. list customers who have placed orders but made no payments

Query :- select distinct c.customerName
from customers c
join orders o on c.customerNumber = o.customerNumber
left join payments p on c.customerNumber = p.customerNumber
where p.customerNumber is null;

Results :-

```
mysql> select distinct c.customerName
-> from customers c
-> join orders o on c.customerNumber = o.customerNumber
-> left join payments p on c.customerNumber = p.customerNumber
-> where p.customerNumber is null;
Empty set (0.01 sec)
```

53. find total number of products ordered by each customer

Query :- select c.customerName, sum(od.quantityOrdered) as total_products,
sum(od.quantityOrdered * od.priceEach) as total_amount
from customers c
join orders o on c.customerNumber = o.customerNumber
join orderdetails od on o.orderNumber = od.orderNumber
group by c.customerName;

Results :-

```
mysql> select c.customerName, sum(od.quantityOrdered) as total_products,
-> sum(od.quantityOrdered * od.priceEach) as total_amount
-> from customers c
-> join orders o on c.customerNumber = o.customerNumber
-> join orderdetails od on o.orderNumber = od.orderNumber
-> group by c.customerName;
+-----+-----+-----+
| customerName | total_products | total_amount |
+-----+-----+-----+
| ABC Corp     | 15             | 375.00       |
| Global Inc   | 3              | 90.00        |
| XYZ Ltd      | 7              | 196.00       |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

54. show product-wise revenue and total quantity sold

Query :- select p.productName,
sum(od.quantityOrdered) as total_qty,
sum(od.quantityOrdered * od.priceEach) as total_revenue
from products p
join orderdetails od on p.productCode = od.productCode
group by p.productName
order by total_revenue desc;

Results :-

```
mysql> select p.productName,
-> sum(od.quantityOrdered) as total_qty,
-> sum(od.quantityOrdered * od.priceEach) as total_revenue
-> from products p
-> join orderdetails od on p.productCode = od.productCode
-> group by p.productName
-> order by total_revenue desc;
```

productName	total_qty	total_revenue
Toyota Camry	13	390.00
Honda Civic	7	196.00
Yamaha Bike	5	75.00

3 rows in set (0.00 sec)

55. find employees along with total sales generated

Query :- select e.firstName, e.lastName,
sum(od.quantityOrdered * od.priceEach) as total_sales
from employees e
join customers c on e.employeeNumber = c.salesRepEmployeeNumber
join orders o on c.customerNumber = o.customerNumber
join orderdetails od on o.orderNumber = od.orderNumber
group by e.employeeNumber;

Results :-

```
mysql> select e.firstName, e.lastName,
-> sum(od.quantityOrdered * od.priceEach) as total_sales
-> from employees e
-> join customers c on e.employeeNumber = c.salesRepEmployeeNumber
-> join orders o on c.customerNumber = o.customerNumber
-> join orderdetails od on o.orderNumber = od.orderNumber
-> group by e.employeeNumber;
```

firstName	lastName	total_sales
Jane	Doe	465.00
Mike	Brown	196.00

2 rows in set (0.00 sec)

56. find customers whose total purchase amount is greater than average

Query :-

Results :-

57. list products ordered in more than 2 different orders

Query :- select productCode, count(distinct orderNumber) as total_orders
from orderdetails
group by productCode
having count(distinct orderNumber) > 2;

Results :-

```
mysql> select productCode, count(distinct orderNumber) as total_orders
-> from orderdetails
-> group by productCode
-> having count(distinct orderNumber) > 2;
Empty set (0.00 sec)
```

58. find customers who ordered more than 3 different products

Query :- select o.customerNumber, count(distinct od.productCode) as total_products
from orders o
join orderdetails od on o.orderNumber = od.orderNumber
group by o.customerNumber
having count(distinct od.productCode) > 3;

Results :-

```
mysql> select o.customerNumber, count(distinct od.productCode) as total_products
-> from orders o
-> join orderdetails od on o.orderNumber = od.orderNumber
-> group by o.customerNumber
-> having count(distinct od.productCode) > 3;
Empty set (0.02 sec)
```

59. show product lines where total revenue exceeds 500

Query :- select p.productLine, sum(od.quantityOrdered * od.priceEach) as total_revenue
from products p
join orderdetails od on p.productCode = od.productCode
group by p.productLine
having total_revenue > 500;

Results :-

```
mysql> select p.productLine, sum(od.quantityOrdered * od.priceEach) as total_revenue
-> from products p
-> join orderdetails od on p.productCode = od.productCode
-> group by p.productLine
-> having total_revenue > 500;
+-----+-----+
| productLine | total_revenue |
+-----+-----+
| Cars        |          586.00 |
+-----+-----+
1 row in set (0.00 sec)
```

60. find countries where average credit limit > 60000

Query :- select country, avg(creditLimit) as avg_credit
from customers
group by country
having avg_credit > 60000;

Results :-

```
mysql> select country, avg(creditLimit) as avg_credit
-> from customers
-> group by country
-> having avg_credit > 60000;
+-----+-----+
| country | avg_credit |
+-----+-----+
| UK      | 70000.000000 |
+-----+-----+
1 row in set (0.01 sec)
```

61. show customer name, order number, product name, and quantity ordered for all orders

Query :- select c.customerName, o.orderNumber, p.productName, od.quantityOrdered
from customers c
join orders o on c.customerNumber = o.customerNumber
join orderdetails od on o.orderNumber = od.orderNumber
join products p on od.productCode = p.productCode;

Results :-

```
mysql> select c.customerName, o.orderNumber, p.productName, od.quantityOrdered
-> from customers c
-> join orders o on c.customerNumber = o.customerNumber
-> join orderdetails od on o.orderNumber = od.orderNumber
-> join products p on od.productCode = p.productCode;
+-----+-----+-----+-----+
| customerName | orderNumber | productName | quantityOrdered |
+-----+-----+-----+-----+
| ABC Corp     | 3001        | Toyota Camry | 10              |
| ABC Corp     | 3001        | Yamaha Bike  | 5               |
| XYZ Ltd      | 3002        | Honda Civic  | 7               |
| Global Inc   | 3003        | Toyota Camry | 3               |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

62. find employee name, office city, and number of customers handled

Query :- select e.firstName, e.lastName, o.city, count(c.customerNumber) as total_customers
from employees e
join offices o on e.officeCode = o.officeCode
left join customers c on e.employeeNumber = c.salesRepEmployeeNumber
group by e.employeeNumber;

Results :-

```
mysql> select e.firstName, e.lastName, o.city, count(c.customerNumber) as total_customers
-> from employees e
-> join offices o on e.officeCode = o.officeCode
-> left join customers c on e.employeeNumber = c.salesRepEmployeeNumber
-> group by e.employeeNumber;
+-----+-----+-----+-----+
| firstName | lastName | city    | total_customers |
+-----+-----+-----+-----+
| John      | Smith   | New York | 0              |
| Jane      | Doe     | New York | 2              |
| Mike      | Brown   | London  | 1              |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

63. display product line, product name, and total revenue generated

Query :- select p.productLine, p.productName, sum(od.quantityOrdered * od.priceEach) as total_revenue
from products p
join orderdetails od on p.productCode = od.productCode
group by p.productCode;

Results :-

```
mysql> select p.productLine, p.productName, sum(od.quantityOrdered * od.priceEach) as total_revenue
-> from products p
-> join orderdetails od on p.productCode = od.productCode
-> group by p.productCode;
```

productLine	productName	total_revenue
Cars	Toyota Camry	390.00
Cars	Honda Civic	196.00
Bikes	Yamaha Bike	75.00

3 rows in set (0.00 sec)

64. show order number, customer name, and total order value

Query :- select o.orderNumber, c.customerName, sum(od.quantityOrdered * od.priceEach) as total_value
from orders o
join customers c on o.customerNumber = c.customerNumber
join orderdetails od on o.orderNumber = od.orderNumber
group by o.orderNumber;

Results :-

```
mysql> select o.orderNumber, c.customerName, sum(od.quantityOrdered * od.priceEach) as total_value
-> from orders o
-> join customers c on o.customerNumber = c.customerNumber
-> join orderdetails od on o.orderNumber = od.orderNumber
-> group by o.orderNumber;
```

orderNumber	customerName	total_value
3001	ABC Corp	375.00
3002	XYZ Ltd	196.00
3003	Global Inc	90.00

3 rows in set (0.00 sec)

65. list employees whose customers have placed more than 5 orders in total

Query :- select e.employeeNumber, e.firstName, e.lastName
from employees e
join customers c on e.employeeNumber = c.salesRepEmployeeNumber
join orders o on c.customerNumber = o.customerNumber
group by e.employeeNumber
having count(o.orderNumber) > 5;

Results :-

```
mysql> select e.employeeNumber, e.firstName, e.lastName
-> from employees e
-> join customers c on e.employeeNumber = c.salesRepEmployeeNumber
-> join orders o on c.customerNumber = o.customerNumber
-> group by e.employeeNumber
-> having count(o.orderNumber) > 5;
Empty set (0.00 sec)
```

Query :-

Query :-

Query :- select c.customerName

```
left join orders o on c.customerNumber = o.customerNumber
```

where o.orderNumber is null;

```
mysql> select c.customerName
-> from customers c
-> left join orders o on c.customerNumber = o.customerNumber
-> join payments p on c.customerNumber = p.customerNumber
-> where o.orderNumber is null;
Empty set (0.00 sec)
```

Query :- select *

```
left join customers c on e.employeeNumber = c.salesRepEmployeeNumber
```

where c.customerNumber is null;

```
mysql> select *
    -> from employees e
    -> left join customers c on e.employeeNumber = c.salesRepEmployeeNumber
    -> where c.customerNumber is null;
```

employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle	customerNumber	customerName	contactLastName	contactFirstName	phone	addressLine1	addressLine2	city	state	postalCode	country	salesRepEmployeeNumber	creditLimit
1001	Smith	John	x191	johnc@company.com	1	NULl	Manager	NULl	NULl	NULl	NULl	NULl	NULl	NULl	NULl	NULl	NULl	NULl	NULl	NULl

```
1 row in set (0.00 sec)

mysql> |
```

70. find orders that contain more than 1 product (multi-line orders)

Query :- select orderNumber, count(productCode) as total_products
from orderdetails
group by orderNumber
having count(productCode) > 1;

Results :-

```
mysql> select orderNumber, count(productCode) as total_products
-> from orderdetails
-> group by orderNumber
-> having count(productCode) > 1;
+-----+-----+
| orderNumber | total_products |
+-----+-----+
| 3001 | 2 |
+-----+-----+
1 row in set (0.00 sec)
```

71. find top 3 customers based on total order value

Query :- select c.customerName, sum(od.quantityOrdered * od.priceEach) as total_value
from customers c
join orders o on c.customerNumber = o.customerNumber
join orderdetails od on o.orderNumber = od.orderNumber
group by c.customerName
order by total_value desc
limit 3;

Results :-

```
mysql> select c.customerName, sum(od.quantityOrdered * od.priceEach) as total_value
-> from customers c
-> join orders o on c.customerNumber = o.customerNumber
-> join orderdetails od on o.orderNumber = od.orderNumber
-> group by c.customerName
-> order by total_value desc
-> limit 3;
+-----+-----+
| customerName | total_value |
+-----+-----+
| ABC Corp | 375.00 |
| XYZ Ltd | 196.00 |
| Global Inc | 90.00 |
+-----+-----+
3 rows in set (0.00 sec)
```

72. show top 5 products based on total revenue

Query :- select p.productName, sum(od.quantityOrdered * od.priceEach) as total_revenue
from products p
join orderdetails od on p.productCode = od.productCode
group by p.productName
order by total_revenue desc
limit 5;

Results :-

```
mysql> select p.productName, sum(od.quantityOrdered * od.priceEach) as total_revenue
-> from products p
-> join orderdetails od on p.productCode = od.productCode
-> group by p.productName
-> order by total_revenue desc
-> limit 5;
```

productName	total_revenue
Toyota Camry	390.00
Honda Civic	196.00
Yamaha Bike	75.00

```
3 rows in set (0.00 sec)
```

73. list top 2 employees based on sales generated

Query :-

Results :-

74. find top 3 countries by number of customers

Query :- select country, count(*) as total_customers

from customers

group by country

order by total_customers desc

limit 3;

Results :-

```
mysql> select country, count(*) as total_customers
-> from customers
-> group by country
-> order by total_customers desc
-> limit 3;
```

country	total_customers
USA	1
UK	1
India	1

```
3 rows in set (0.00 sec)
```

75. show top 5 most expensive products based on msrp

Query :- select productName, msrp

from products

order by msrp desc

limit 5;

Results :-

```
mysql> select productName, msrp
-> from products
-> order by msrp desc
-> limit 5;
```

productName	msrp
Toyota Camry	30.00
Honda Civic	28.00
Yamaha Bike	15.00

3 rows in set (0.00 sec)

76. find average order value per customer and list only those above overall average

Query :-

Results :-

77. show product-wise average price sold vs msrp difference

Query :-

Results :-

78. find percentage contribution of each product to total revenue

Query :-

Results :-

79. calculate average quantity ordered per order for each customer

Query :-

Results :-

80. find customers whose largest single order value exceeds 10000

Query :-

Results :-